International Institute of Medical Science and Technology council

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A

Mini Project Report

On

"AQUAGUARD COMMUNITY MONITORING"

(SDG 6 – Clean Water and Sanitation)

Submitted By

Cohort No: 05

ABSTRACT

Problem Statement

Water quality monitoring in many communities is still dependent on manual reporting systems. Citizens must visit offices or make phone calls to register complaints. This process is slow, lacks transparency, and provides no proper tracking mechanism for submitted complaints. There is also limited communication between citizens and authorities, leading to delayed responses and unresolved issues.

Solution – AquaGuard System

AquaGuard is a full-stack web-based platform designed to enable community-driven water quality reporting. It allows users to submit water-related complaints along with location details and supporting evidence. Moderators and authorities can review, validate, and manage reports through a centralized dashboard.

Technologies Used

- React.js (Frontend)
- Node.js & Express.js (Backend)
- MongoDB (Database)
- Google Maps API (Location tracking)
- JWT (Authentication)
- Cloud Storage (File upload)

Outcome

The system improves transparency, enables real-time tracking of complaints, reduces response time, and strengthens communication between community members and authorities.

CHAPTER 1: INTRODUCTION

1.1 Background

Water quality monitoring is essential to ensure safe drinking water and environmental protection. In many regions, water contamination issues such as pollution, sewage mixing, and chemical discharge are common. Traditional monitoring systems depend heavily on manual inspections and citizen complaints.

Manual systems suffer from:

- Delayed complaint registration
- Lack of digital record keeping
- No real-time tracking
- Poor transparency
- Inefficient communication

There is a need for a digital platform that enables quick reporting, monitoring, and management of water-related issues.

1.2 Problem Statement

The existing manual complaint system for water-related issues is inefficient and lacks transparency. Citizens face difficulties in reporting problems, and there is no proper tracking mechanism to monitor complaint status. Authorities cannot easily verify reports or maintain structured data. This results in delayed action and reduced public trust.

1.3 Objectives

The main objectives of AquaGuard are:

- To enable community-based reporting of water quality issues.
- To provide a real-time dashboard for monitoring complaints.
- To validate and manage reports through moderators.
- To notify authorities and users about report status updates.

1.4 Scope of the Project

The scope of AquaGuard includes:

- Web-based complaint submission system
- Role-based access (Community Members and Moderators)
- Location-based issue tracking using maps
- Secure authentication and authorization
- File upload for evidence
- Real-time status updates

The system is designed for community-level deployment and can be extended to municipal or district-level use.

CHAPTER 2: LITERATURE SURVEY

Existing Water Monitoring Systems

Several water monitoring systems exist that use IoT sensors and laboratory testing methods to detect contamination levels. However, many of these systems focus only on technical data collection and lack community participation features.

Most traditional systems:

- Do not allow citizen reporting
- Lack centralized digital dashboards
- Provide limited transparency

Government Reporting Applications

Some government applications allow citizens to submit complaints regarding civic issues. However:

- Many apps are not user-friendly.
- Tracking mechanisms are limited.
- Notifications are inconsistent.
- Data visualization is minimal.

AquaGuard improves upon these systems by integrating community reporting, map visualization, dashboard analytics, and automated notifications.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Existing System

The existing system involves:

- Manual complaint submission
- Phone calls or physical visits
- Paper-based records
- No real-time tracking
- Lack of transparency

Problems include slow response time and poor accountability.

3.2 Proposed System

The proposed AquaGuard system is a community-driven digital platform that:

- Allows online complaint submission
- Stores reports in a centralized database
- Displays issues on a map
- Enables moderators to validate reports
- Sends automated notifications

This system improves efficiency, transparency, and tracking.

3.3 Feasibility Study

Technical Feasibility

The project uses widely available technologies such as React, Node.js, and MongoDB. These tools are stable, scalable, and suitable for web application development.

Economic Feasibility

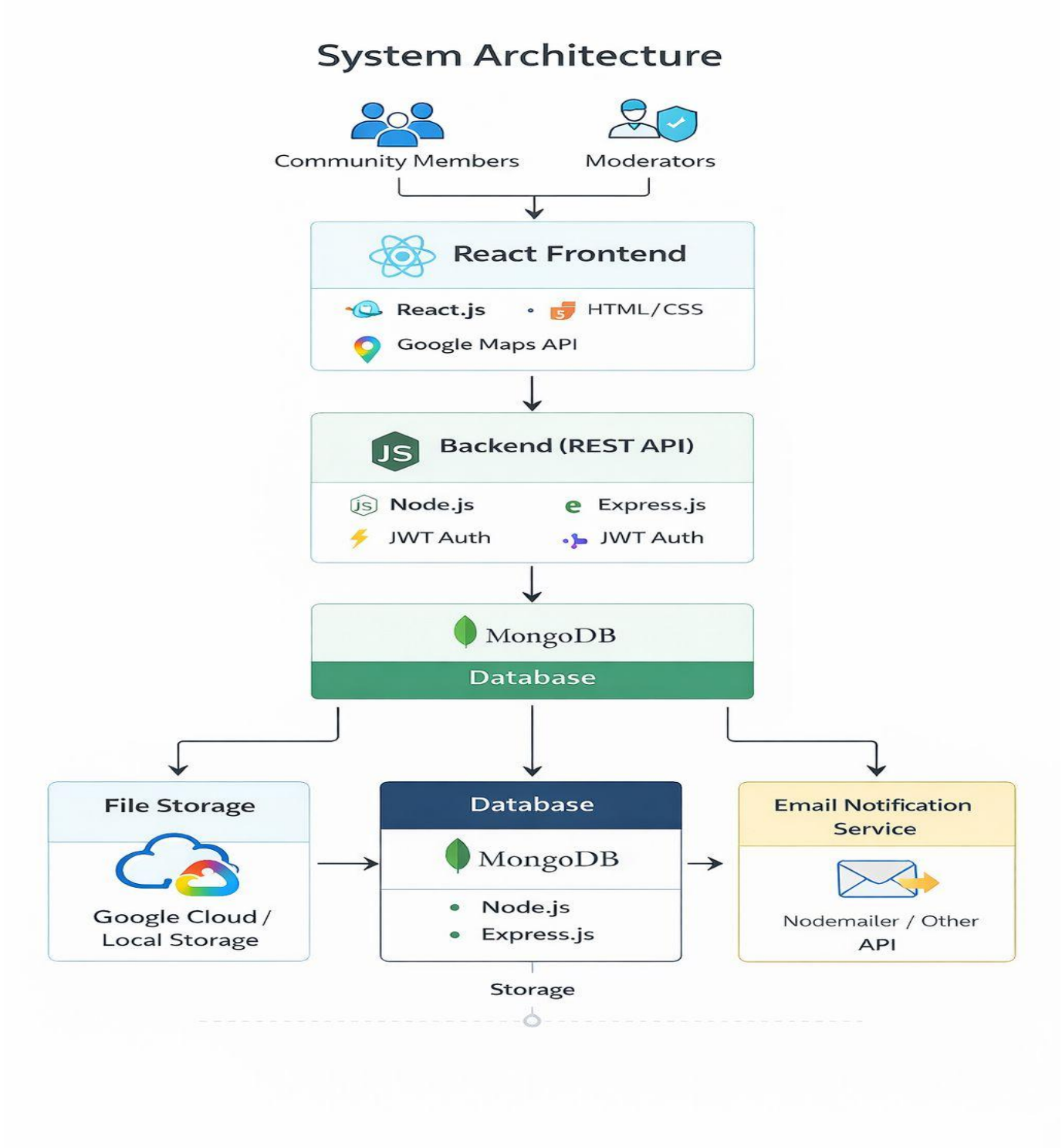
The system uses open-source technologies, reducing development costs. Cloud hosting costs are minimal and scalable based on usage.

Operational Feasibility

The system is user-friendly and requires minimal training. Community members can easily submit reports through a simple interface.

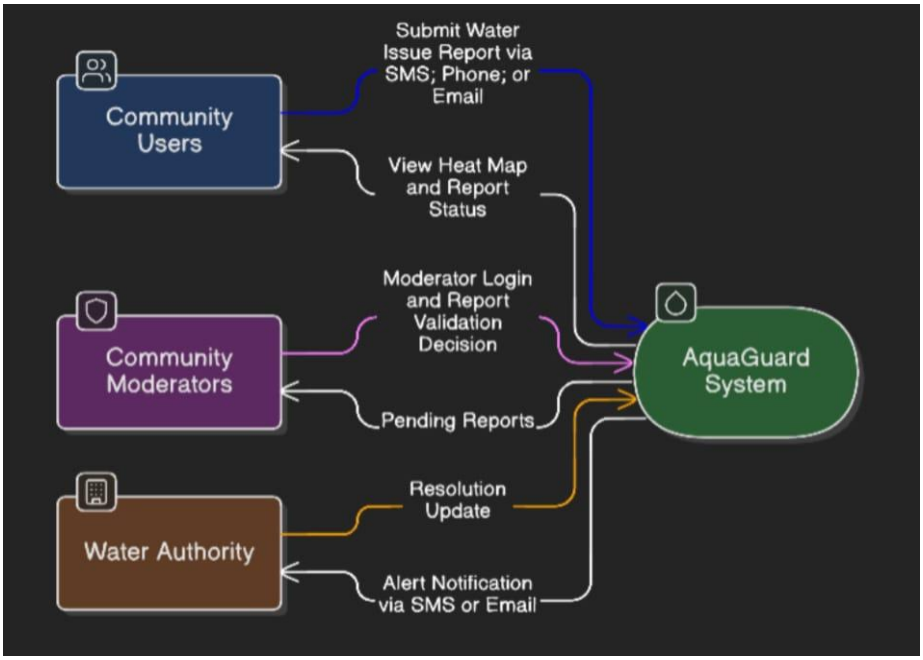
CHAPTER 4: SYSTEM DESIGN

4.1 Overall System Architecture

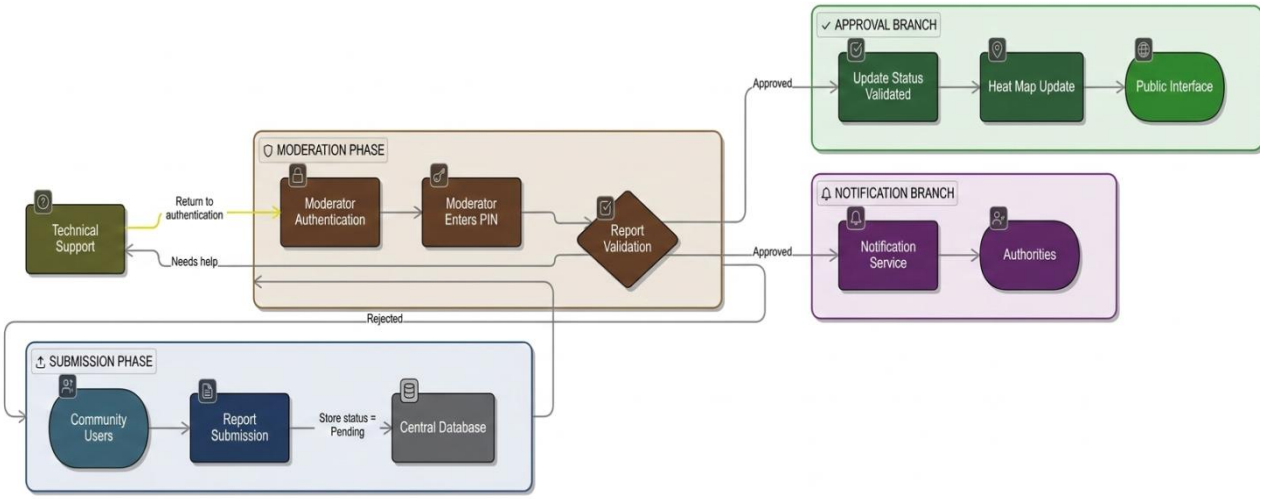


4.2 Data Flow Diagram

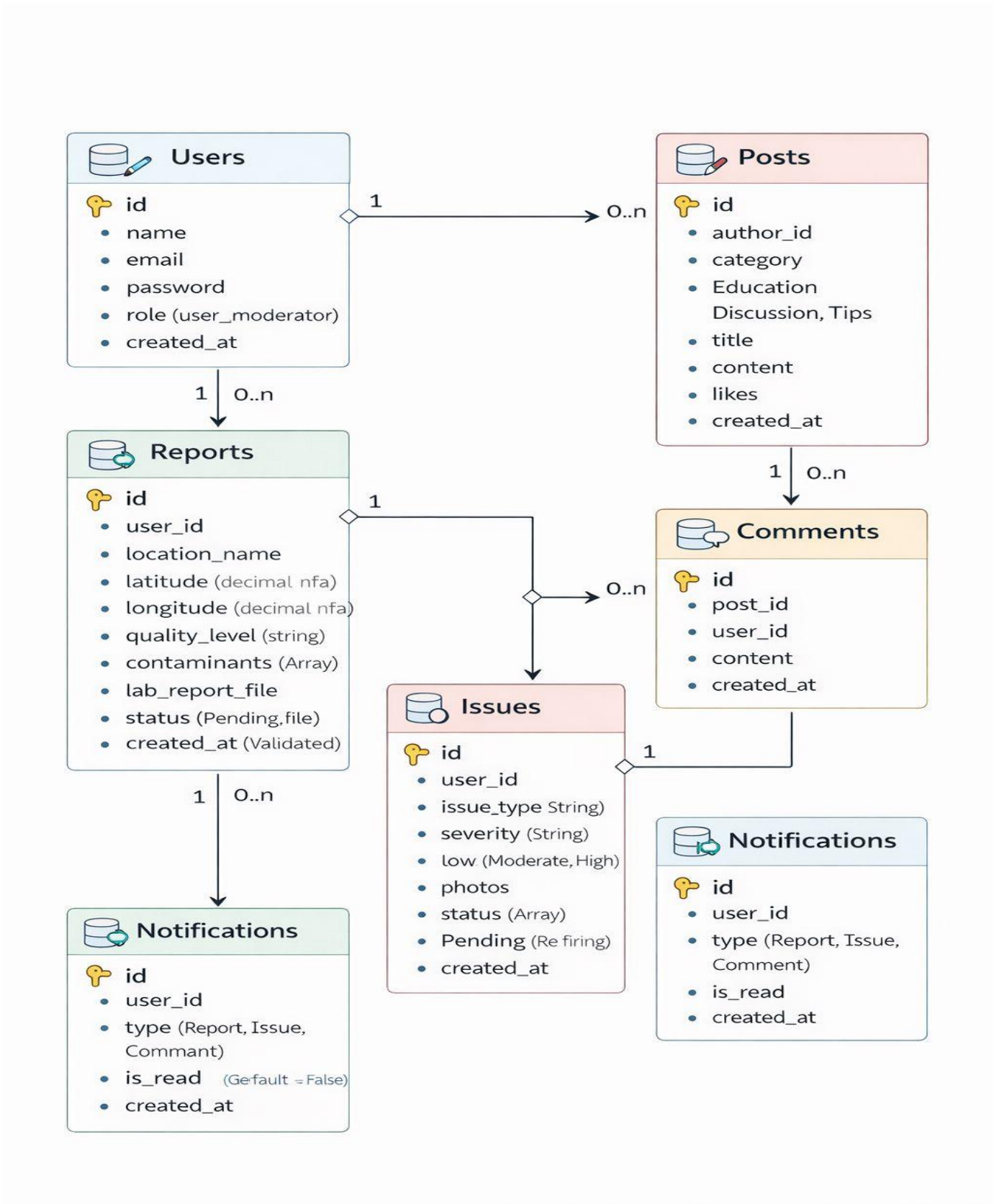
DFD Level 0



DFD Level 1



4.3 Database Design



CHAPTER 5: IMPLEMENTATION

5.1 Frontend Implementation

Developed using React.js with components such as:

- Login Component
- Registration Component
- Dashboard Component
- Report Submission Form
- Moderator Panel
- Map Component

5.2 Backend Implementation

Built using Express APIs:

- /register
- /login
- /submit-report
- /get-reports
- /update-status

JWT is used for secure authentication.

5.3 Database Implementation

MongoDB schemas are defined for:

- Users
- Reports
- Notifications

Mongoose is used to connect Node.js with MongoDB.

5.4 API Endpoints

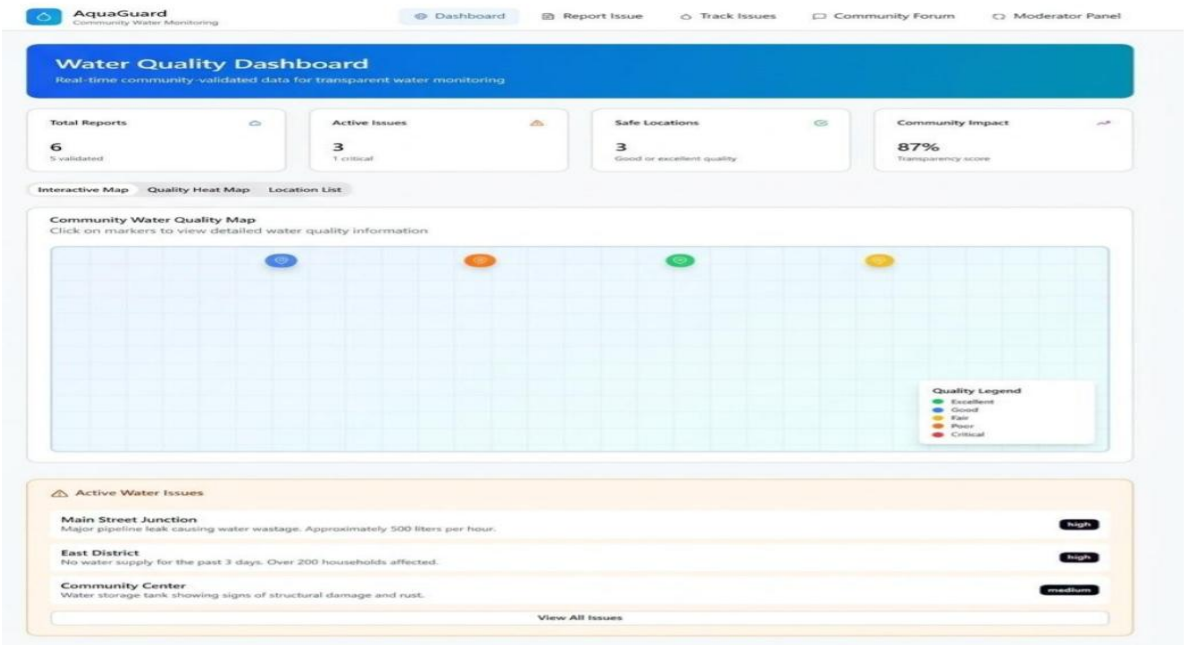
Examples:

POST /api/register
POST /api/login
POST /api/report
GET /api/reports
PUT /api/report/:id

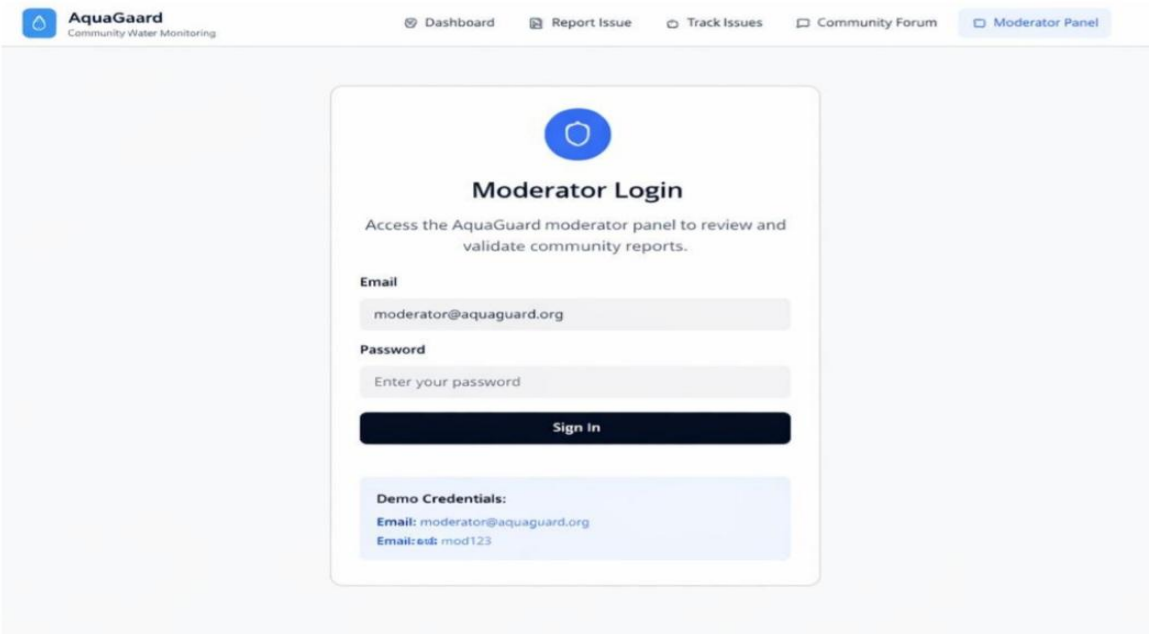
CHAPTER 6: RESULTS AND OUTPUT

The system successfully provides:

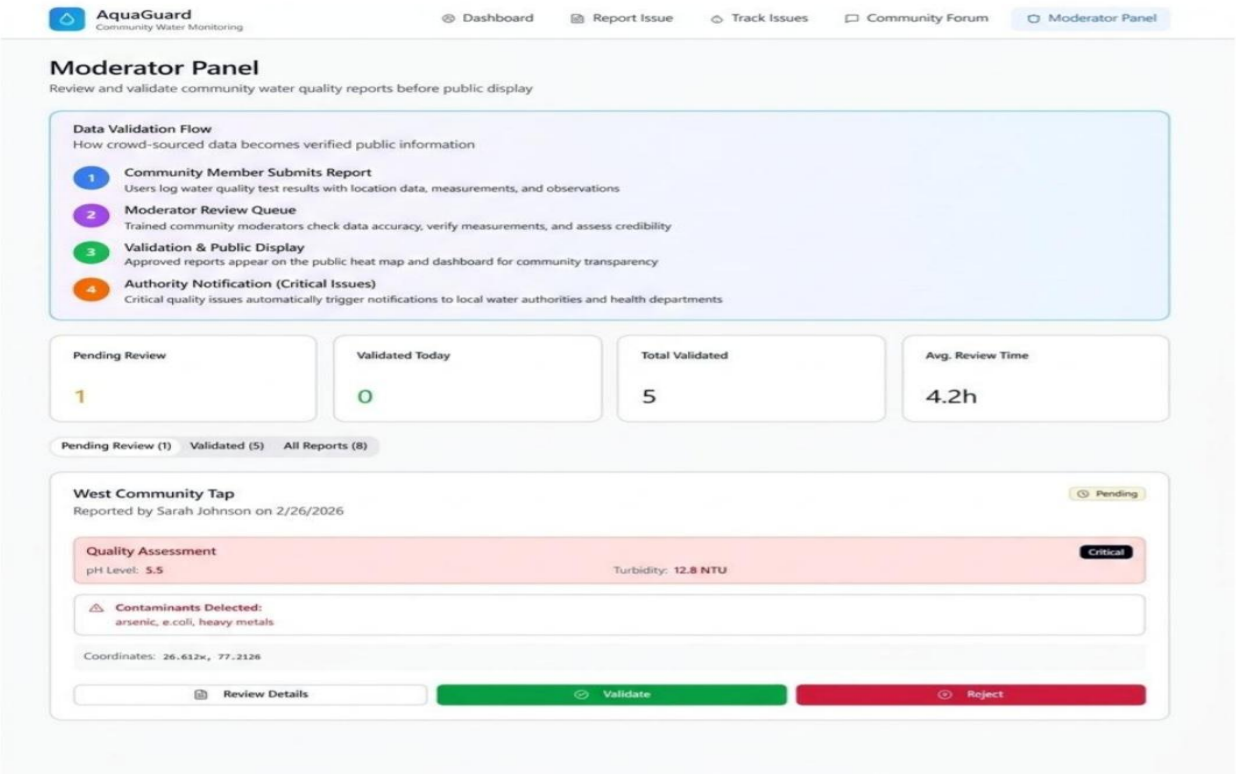
6.1 User Dashboard



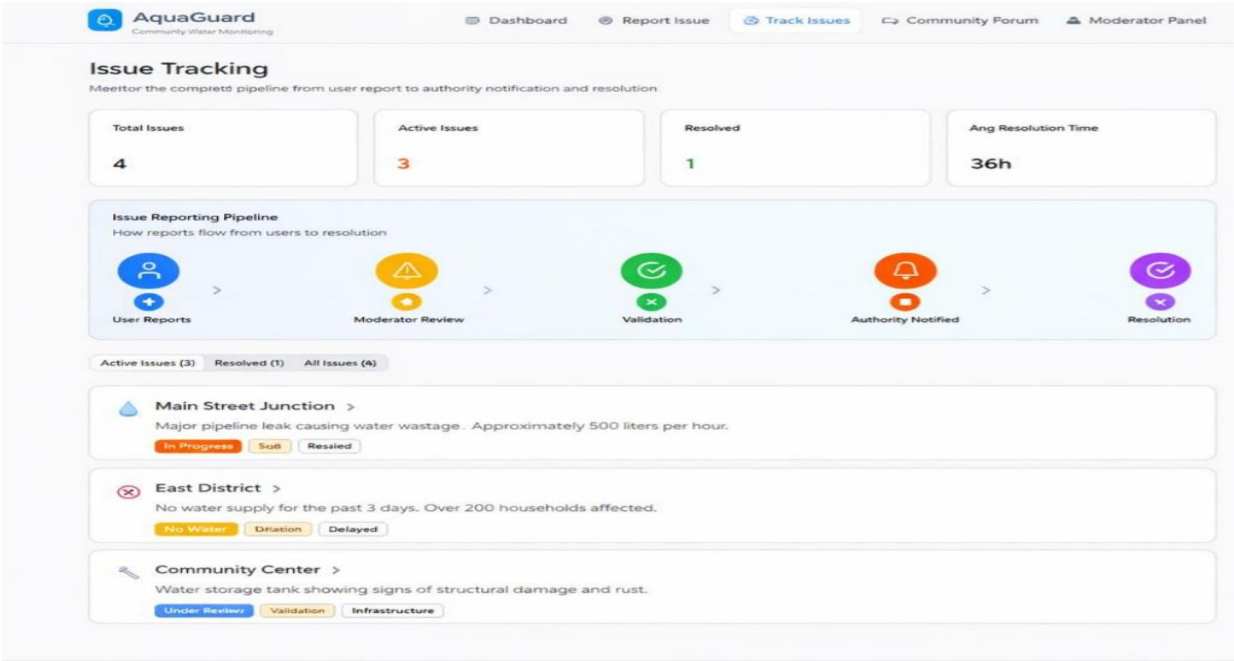
6.2 Logging Page



6.3 Moderator Panel



6.4 Issue tracking page



CHAPTER 7: SDG Impact Analysis

SDG 7.1: Clean Water and Sanitation

Goal Description:

Ensure availability and sustainable management of water and sanitation for all.

How AquaGuard Supports SDG 7.1:

- Enables citizens to report water contamination issues.
- Provides location-based tracking of polluted water sources.
- Helps authorities identify high-risk areas.
- Encourages community participation in water quality monitoring.
- Promotes faster response to unsafe water conditions.

Impact:

AquaGuard strengthens local water governance by making contamination reporting transparent and trackable, thereby improving public health and water safety.

SDG 7.2: Sustainable Cities and Communities

Goal Description:

Make cities inclusive, safe, resilient, and sustainable.

Contribution of AquaGuard:

- Digital complaint management improves civic infrastructure monitoring.
- Real-time dashboards help track environmental issues.

- Promotes accountability of local authorities.
- Encourages citizen engagement in urban management.

Impact:

The platform builds a smarter and more responsive community ecosystem by digitizing civic issue reporting.

SDG 7.3: Peace, Justice and Strong Institutions

Goal Description:

Promote peaceful and inclusive societies and build accountable institutions.

AquaGuard Contribution:

- Provides transparent tracking of complaints.
- Reduces corruption through digital records.
- Ensures traceability of report handling.
- Implements role-based moderation for accountability.

Impact:

The system strengthens institutional transparency and promotes trust between citizens and authorities.

SDG 7.4: Industry, Innovation and Infrastructure

Goal Description:

Build resilient infrastructure and foster innovation.

AquaGuard's Role:

- Implements modern MERN-stack architecture.
- Uses cloud storage for scalable infrastructure.
- Integrates map-based visualization.
- Promotes digital transformation of civic services.

Impact:

Encourages innovation in environmental governance through digital technology.

CHAPTER 8: CONCLUSION

The AquaGuard system has been successfully designed and implemented as a full-stack web application. It provides a transparent and efficient platform for water quality issue reporting. The system enhances community participation and improves communication between citizens and authorities.

The project achieves its objective of digitalizing water complaint management and ensuring transparency.

CHAPTER 9: FUTURE ENHANCEMENTS

Future improvements include:

- AI-based water contamination prediction
- IoT sensor integration for real-time water monitoring
- SMS alert system
- Mobile application version
- Advanced analytics dashboard